Data Communication Basics

Q: What is data communication?
A: Data communication is the exchange of data between two devices via a transmission medium.

Q: What are the four fundamental characteristics of data communication?
A: Delivery, accuracy, timeliness, and jitter.

Q: What does "delivery" mean in data communication?
A: Data must reach the correct and intended destination only.

Q: Why is accuracy important in data communication?
A: Altered or corrupted data are unusable and must be avoided.

Q: What is timeliness in data communication?
A: Data must be delivered in real time, in order, and without significant delay.

Q: What is jitter?
A: Jitter is the variation in packet arrival time, causing uneven audio/video quality.

Components of Data Communication

Q: Name the five components of a data communication system.
A: Message, sender, receiver, transmission medium, and protocol.

Q: What is a protocol?
A: A protocol is a set of rules governing data communication between devices.

Data Representation

Q: How is text represented in data communication?
A: As bit patterns using codes like ASCII (7-bit), Extended ASCII (8-bit), or Unicode (16-bit).

Q: How are numbers represented?
A: Directly converted into binary form.

Q: Name common formats for audio and video.
A: Audio: MP3, WMA; Video: MP4, AVI.

Data Flow

Q: What is simplex data flow?
A: Communication is unidirectional; one device sends, the other only receives.

Q: Give an example of simplex communication.
A: Keyboard (sends only) and traditional monitor (receives only).

Q: What is half-duplex communication?
A: Both devices can send and receive, but not at the same time.

Q: Give an example of half-duplex communication.
A: Walkie-talkies or CB radios.

Q: What is full-duplex communication?
A: Both devices can transmit and receive simultaneously.

Q: Give an example of full-duplex communication.
A: Telephone conversation.

Network Criteria

Q: What are the three main criteria of a network?
A: Performance, reliability, and security.

Q: How is network performance measured?
A: By transit time, response time, throughput, and delay.

Q: What is transit time?
A: Time taken for a message to travel from sender to receiver.

Q: What is response time?
A: Time between an inquiry and its response.

Q: How is network reliability measured?
A: By frequency of failure, recovery time, and robustness during catastrophes.

Q: What does network security involve?
A: Protecting data from unauthorized access, damage, and ensuring recovery from breaches.

Types of Connection

Q: What is a point-to-point connection?
A: A dedicated link between two devices.

Q: What is a multipoint connection?
A: More than two devices share a single link.

Physical Topologies (slide)

Q: How many links are in a fully connected mesh network with *n* devices?
A: duplex links.

Q: What is an advantage of mesh topology?
A: Dedicated links ensure no traffic congestion and high privacy.

Q: What is a disadvantage of mesh topology?
A: Expensive due to high cabling and I/O port requirements.

Q: How does star topology work?
A: All devices connect to a central hub, which relays data.

Q: What happens if the hub fails in a star topology?
A: The entire network goes down.

Q: What is a bus topology?
A: All devices connect to a single backbone cable via drop lines and taps.

Q: What is a major drawback of bus topology?
A: A break in the cable stops all communication.

Q: How does data travel in a ring topology?
A: In one direction, with each device acting as a repeater.

Q: What is a hybrid topology?
A: A combination of two or more topologies, e.g., star backbone with bus branches.

Network Categories

Q: What is a LAN?
A: A privately owned network covering a small area like a building or campus.

Q: What is the typical range of a LAN?
A: 1 to 10 km.

Q: What is a MAN?
A: A network spanning a city (around 100 km), e.g., DSL or cable TV networks.

Q: What is a WAN?
A: A network covering large geographic areas, even globally.

Q: Which network uses only guided media?
A: LAN.

Q: Which network is most expensive?
A: WAN.

Protocols

Q: What are the three key elements of a protocol?
A: Syntax, semantics, and timing.

Q: What is syntax in a protocol?
A: The structure or format of data (e.g., order of bits).

Q: What is semantics in a protocol?
A: The meaning of each bit pattern and the action to be taken.

Q: What is timing in a protocol?
A: When to send data and how fast to send it.